PRACTICAL NO. 7

Design and Implement Dynamic Routing Using GNS3 software

Aim:

To design and configure **dynamic routing using EIGRP (Enhanced Interior Gateway Routing Protocol)** in GNS3 with **2 routers and 2 PCs**, and verify communication between the two different networks.

Theory:

EIGRP (Enhanced Interior Gateway Routing Protocol) is a dynamic routing protocol developed by Cisco. It is an advanced distance-vector routing protocol that uses the **DUAL (Diffusing Update Algorithm)** to calculate the best path to a destination.

EIGRP uses a composite metric primarily based on **bandwidth and delay**. It allows routers to automatically learn remote networks and update their routing tables when there is a change in the network.

Advantages of EIGRP

- Fast convergence
- Supports VLSM and CIDR
- Uses bandwidth and delay for route selection
- Maintains neighboring router information
- Supports automatic route summarization
- Suitable for medium and large networks

Device	Interface	IP Address	Subnet Mask
PC1	Ethernet	192.168.1.10	255.255.255.0
R1	G0/0	192.168.1.1	255.255.255.0
R1	G0/1	10.0.0.1	255.255.255.252
R2	G0/1	10.0.0.2	255.255.255.252
R2	G0/0	192.168.2.1	255.255.255.0
PC2	Ethernet	192.168.2.10	255.255.255.0

Procedure

Step 1: Create the Topology in GNS3

1. Open **GNS3**.
2. Create a new project.
3. Add:
    1. 2 routers — R1 and R2
    2. 2 VPCS — PC1 and PC2
4. Connect the devices:

PC1 ---- R1 ---- R2 ---- PC2

1. Start all devices.

Step 2: Configure Router R1

Open the console of R1.

Configure the interfaces

```
enable
configure terminal

hostname R1

interface g0/0
ip address 192.168.1.1 255.255.255.0
no shutdown
exit

interface g0/1
ip address 10.0.0.1 255.255.255.252
no shutdown
exit
```

Configure EIGRP

Use an **EIGRP autonomous system number**, for example 100.

```
router eigrp 100
no auto-summary
network 192.168.1.0 0.0.0.255
network 10.0.0.0 0.0.0.3
exit
```

end
write memory

Step 3: Configure Router R2

Open the console of R2.

Configure the interfaces

enable
configure terminal

hostname R2

interface g0/0
ip address 192.168.2.1 255.255.255.0
no shutdown
exit

interface g0/1
ip address 10.0.0.2 255.255.255.252
no shutdown
exit

Configure EIGRP

router eigrp 100
no auto-summary
network 192.168.2.0 0.0.0.255
network 10.0.0.0 0.0.0.3
exit

end
write memory

Step 4: Configure PC1

Open the PC1 console:

ip 192.168.1.10/24 192.168.1.1

Verify:

show ip

Step 5: Configure PC2

Open the PC2 console:

ip 192.168.2.10/24 192.168.2.1

Verify:

show ip

Step 6: Verify Router Interfaces

On R1:

show ip interface brief

On R2:

show ip interface brief

Step 7: Verify EIGRP Neighbor Relationship

On R1:

show ip eigrp neighbors

Similarly, on R2:

show ip eigrp neighbors

Step 8: Verify the Routing Table

On R1:

show ip route

On R2:

show ip route

Step 9: Verify EIGRP Configuration

Use:

show ip protocols
show ip eigrp topology

Step 10: Test Connectivity

From PC1 to PC2

On PC1:

ping 192.168.2.10

From PC2 to PC1

On PC2:

ping 192.168.1.10

Conclusion:

Thus, **dynamic routing using EIGRP was successfully designed and configured in GNS3 using 2 routers and 2 PCs**. The routers exchanged routing information dynamically through EIGRP, and connectivity between PC1 and PC2 was successfully verified using the ping command.